Lab Exercise 4- Building a Docker Image for an HTML App Using Nginx

1. Setup

You will need:

- Docker installed on your machine.
- A simple HTML file for the app.

2. Step 1: Create the HTML File

Create a directory for your HTML app and place an index.html file in it.

```
mkdir nginx-html-app

cd nginx-html-app
```

Inside the nginx-html-app directory, create the HTML file.

```
touch index.html
```

Edit the index.html file with the following content (or any custom HTML content you want):

```
<!DOCTYPE html>
<html>
<head>
  <title>Welcome to My Nginx HTML App</title>
</head>
<body>
  <h1>Hello, Nginx Docker!</h1>
  <p>This is a simple HTML app served by Nginx in a Docker container.</p>
</body>
</html>
```

3. Step 2: Create a Dockerfile

In the same directory, create a Dockerfile. This file will define how to build the Docker image using Nginx as the base image.

```
touch Dockerfile
```

Edit the Dockerfile and add the following content:

```
FROM nginx:latest
COPY index.html /usr/share/nginx/html/
EXPOSE 80
```

4. Step 3: Build the Docker Image

Now that you have the Dockerfile and index.html, it's time to build the Docker image. Run the following command to build the image, giving it a tag (e.g., nginx-html-app):

```
docker build -t nginx-html-app .
```

Docker will use the Nginx base image, copy your index.html into the appropriate directory, and build the image.

5. Step 4: Run the Docker Container

After building the image, you can run the container with the following command:

```
docker run -d -p 8006:80 nginx-html-app
```

This command runs the container in detached mode (-d) and maps port 8006 on your host machine to port 80 inside the container, where Nginx is serving your HTML app.

6. Step 5: Verify

Open a browser and go to http://localhost:8006. You should see your HTML page with the message "Hello, Nginx Docker!".

7. Step 6: Stop and Remove the Container

Once you're done, you can stop and remove the container:

docker ps # to see running containers

docker stop <container-id>

docker rm <container-id>

```
C:\Users\NK\nginx-html-app>docker run -d -p 8006:80 nginx-html-app
6b8b27c439fef9b1ff60da3e6176c48c9ffc56b782ac4541fc9016b926210949

C:\Users\NK\nginx-html-app>docker ps
CONTAINER ID   IMAGE           COMMAND                  CREATED         STATUS         PORTS                                     NAMES
6b8b27c439fe   nginx-html-app   "/docker-entrypoint.…"   2 minutes ago   Up 2 minutes   0.0.0.0:8006->80/tcp, [::]:8006->80/tcp   distracted_bose

C:\Users\NK\nginx-html-app>docker stop <container-id>
The syntax of the command is incorrect.

C:\Users\NK\nginx-html-app>docker stop <6b8b27c439fe>
The syntax of the command is incorrect.

C:\Users\NK\nginx-html-app>docker stop 6b8b27c439fee
Error response from daemon: No such container: 6b8b27c439fee

C:\Users\NK\nginx-html-app>docker stop 6b8b27c439fe
6b8b27c439fe

C:\Users\NK\nginx-html-app>docker rm 6b8b27c439fe
6b8b27c439fe

C:\Users\NK\nginx-html-app>
```

```
 & echo ^</body^> >> index.html & echo ^</html^> >> index.html

C:\Users\NK\nginx-html-app>echo FROM nginx:latest > Dockerfile & echo COPY index.html /usr/share/nginx/html/ >> Dockerfi
le & echo EXPOSE 80 >> Dockerfile

C:\Users\NK\nginx-html-app>docker build -t nginx-html-app .
[+] Building 1.2s (7/7) FINISHED                                                                  docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                0.1s
 => => transferring dockerfile: 112B                                                                                0.0s
 => [internal] load metadata for docker.io/library/nginx:latest                                                     0.1s
 => [internal] load .dockerignore                                                                                   0.1s
 => => transferring context: 2B                                                                                     0.0s
 => [1/2] FROM docker.io/library/nginx:latest@sha256:c881927c4077710ac4b1da63b83aa163937fb47457950c267d92f7e4dedf  0.2s
 => => resolve docker.io/library/nginx:latest@sha256:c881927c4077710ac4b1da63b83aa163937fb47457950c267d92f7e4dedf  0.0s
 => [internal] load build context                                                                                   0.1s
 => => transferring context: 267B                                                                                   0.0s
 => [2/2] COPY index.html /usr/share/nginx/html/                                                                    0.1s
 => exporting to image                                                                                              0.4s
 => => exporting layers                                                                                             0.1s
 => => exporting manifest sha256:d2dc18e0c7b32610cefcb0dbd78e02a02dc561cca090eae1b5b60a20f7546cb7                   0.0s
 => => exporting config sha256:f7723d4c2f4650a4d48b2d3423e9fbd4f42276da1a0a01e3827e6f997a1f90aa                     0.0s
 => => exporting attestation manifest sha256:f7153aab48349667229fffb4f8cdc461e8860cbb2737c6d919669f285abb7933       0.1s
 => => exporting manifest list sha256:2548991e245e596cf3ad88558ee9e0e3974bf173533d195ec00d6312d0959ed1              0.0s
 => => naming to docker.io/library/nginx-html-app:latest                                                            0.0s
 => => unpacking to docker.io/library/nginx-html-app:latest                                                         0.0s

C:\Users\NK\nginx-html-app>docker run -d -p 8006:80 nginx-html-app
6b8b27c439fef9b1ff60da3e6176c48c9ffc56b782ac4541fc9016b926210949

C:\Users\NK\nginx-html-app>docker ps
```